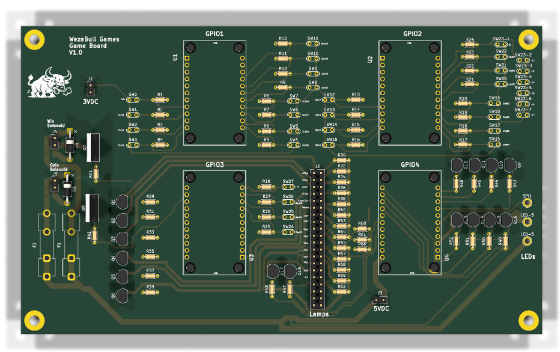
ASSEMBLY GUIDE

SKILLGAME

A TRIBUTE TO THE GAME OF SKILL



ALL HUFF · NO BULL

WezeBull Games · build it once, build it right

Parts **list**

EVERYTHING YOU NEED BEFORE YOU START

QTY	PART	WHAT IT'S FOR
THE BOARD		
4	Adafruit FT232H breakout	USB→GPIO controllers (U1–U4)
16	PN2222A transistor (TO-92)	lamp / reel drivers
2	TIP120 Darlington (TO-220)	solenoid drivers (Q7/Q8)
2	1N4004 diode	solenoid flyback (D1/D2)
28	10 kΩ resistor	switch pull-downs
16	1 kΩ resistor	transistor base
16	0 Ω jumper (or solder bridge)	lamp links R30–R62 — the #555 bulb self-limits
28	100 nF ceramic cap (0805 SMD)	C1–C28 · switch-input debounce/noise (back of board)
2	2.2 kΩ resistor	TIP120 base
3	5×20 mm fuses · 4 Keystone clips + 1 inline holder	F1/F2 T2A (solenoids) · F3 T3.5A inline (5 V lead to J3)
1	2×20 pin header (J2)	lamp / output connector
4	2-pin headers (J1/J3/J4/J5)	power in + solenoid out
~34	Playfield microswitches (hinge lever)	28 inputs (S23 = 7 gobble) · lever pokes up through the slot
1	WS2812b LED strip, 150 px	attract / distract shows
20	#555 wedge LED bulb (6.3 V, non-polar)	16 lamps + a 2nd bulb on 100/200/300/400 (double bright)
20	#555 / T10 wedge socket (twist-lock)	one per bulb; solder its leads to the J2 lamp lines
2	SparkFun Solenoid - 5V (Small)	coin-lock + winner-lock
THE SYSTEM		
1	PELADN Ryzen 5 5600H mini-PC (16 GB / 512 GB)	runs SkillGame
1	QQU 15.6" portable touchscreen (with stand)	the display · 1080P IPS/HDR touch
1	4-port USB 2.0 hub	one port per FT232H
1	3.3 V wall adapter	switch / GPIO logic (J1)

1	5 V wall adapter (≥5 A)	lamps · LEDs · coils (J3 / LED±)

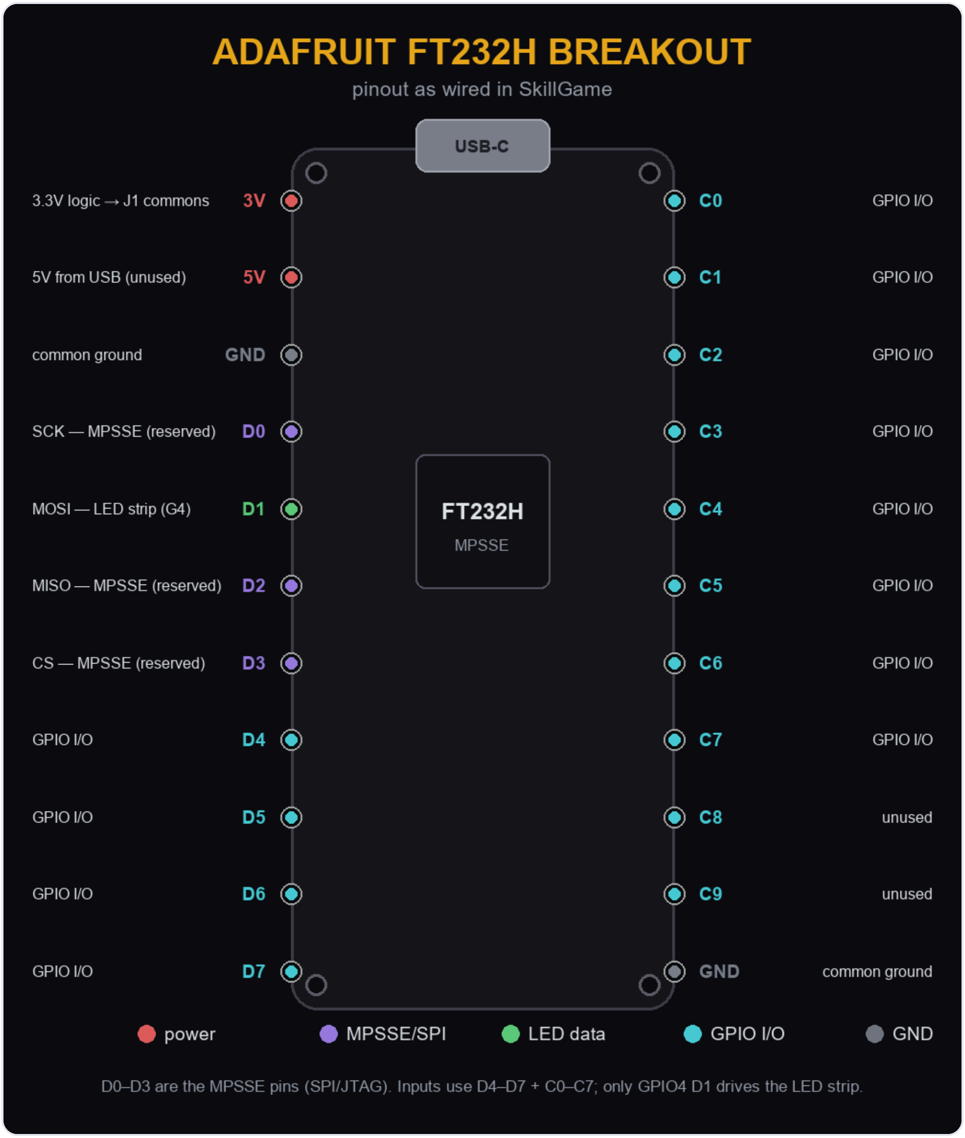
TOOLS

- Soldering iron + solder
- Flush cutters
- Small screwdriver
- Multimeter
- Wire + crimp/ferrules

> **Before you power up:** the 3.3 V adapter goes to J1 (switch commons) and the 5 V to J3 / LED±. Never feed 5 V into J1 — the FT232H inputs are not 5 V-tolerant.

1 Meet the four boards

Adafruit FT232H — one per GPIO bank

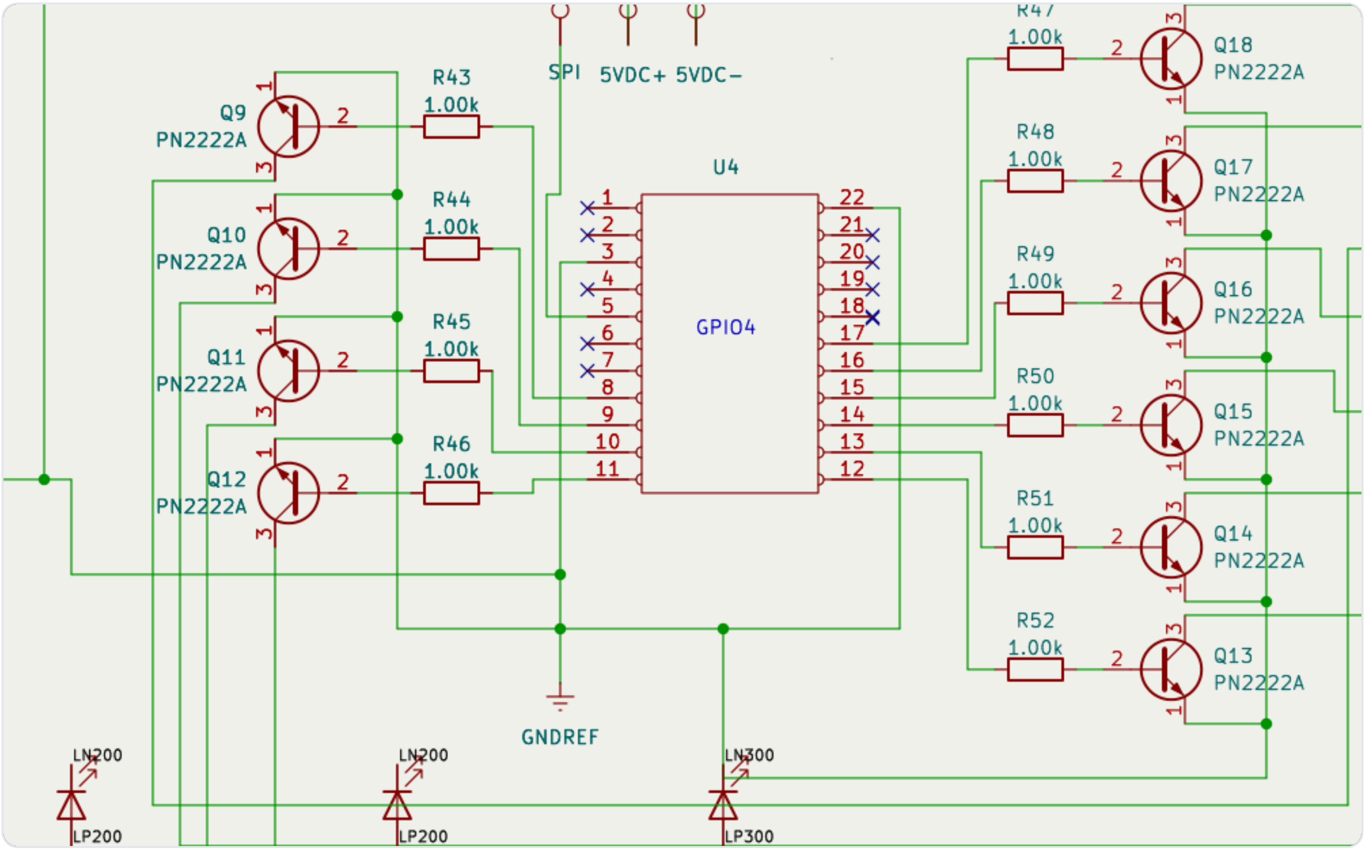


- Four Adafruit FT232H breakouts, each in MPSSE mode over USB-C — the app talks to all four through the USB hub.
- **GPIO1 / GPIO2** read switches; **GPIO3** reads the top rows + tilt and drives the 100–400 reels, Winner, Game-Over and the two coil lines; **GPIO4** drives the 10–90 + Tilt lamps and clocks the LED strip.
- Pins **D0–D3** are reserved (SPI/JTAG) — all switch/lamp I/O lives on **D4–D7** and **C0–C7**. Only GPIO4's D1 is used, for the LED strip.
- Program each board with a unique FTDI serial (GPIO1–GPIO4) so the software binds the right board to the right job no matter which USB port it lands on.

- The FT232H's own 3 V / 5 V pins stay unconnected — the boards are USB-powered.

2 Solder the drivers

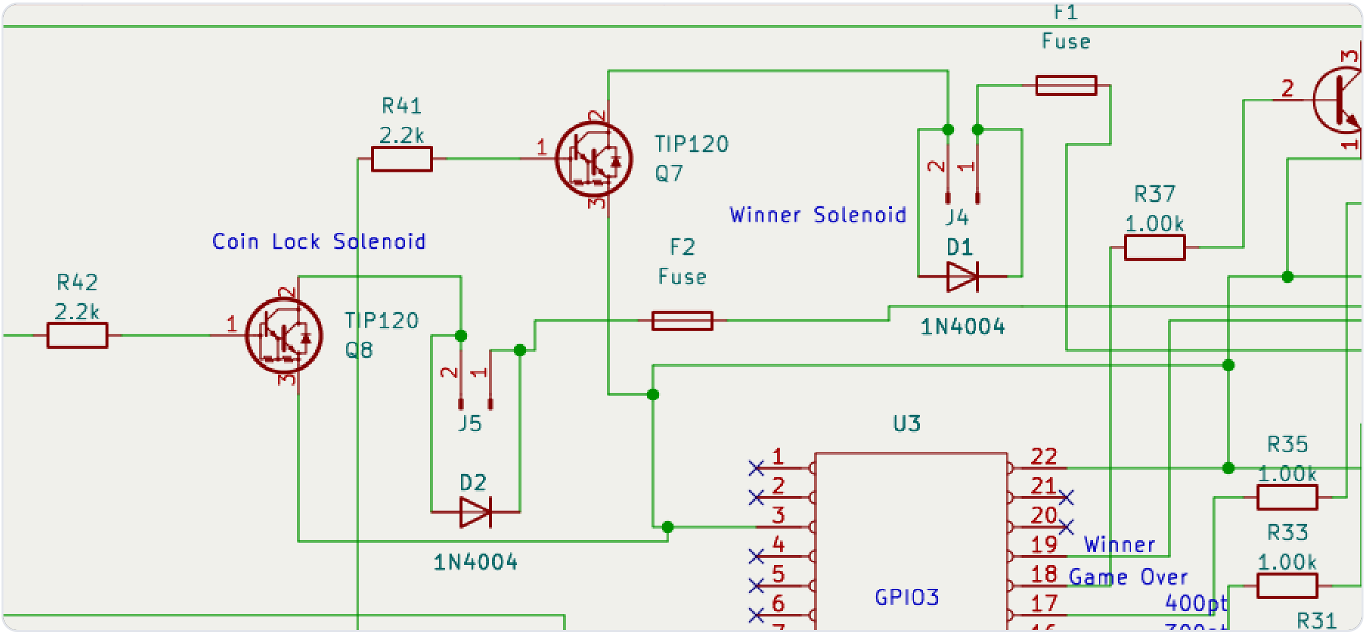
resistors, then transistors — lowest parts first



- Resistors first (they lie flat): 28× 10 kΩ pull-downs, 16× 1 kΩ bases, 2× 2.2 kΩ TIP120 bases. The 16 lamp positions (R30–R62) are **0 Ω links** — a jumper wire or solder bridge, since the #555 bulbs self-limit.
- Then the 16 PN2222A transistors (TO-92) — mind the flat/rounded face matches the silk.
- Keep leads short; flush-cut the backs.

3 Solenoid drivers + fuses

TIP120s, flyback diodes, F1/F2



⚠ **TIP120 orientation:** collector goes to the coil, emitter to ground. If you are using an **older board revision**, cross the collector (middle/tab) and emitter (right) leads on Q7 & Q8 — a straight insert wires them backwards and the coils won't switch. The current KiCad files already have this fixed.

- Fit the two TIP120s (TO-220) and the 1N4004 flyback diodes — band matches the silk.
- Install the Keystone 3517 clips and drop in the fuses (see next page).

3b Fuse the solenoids

F1 / F2 — one per coil, on the board

FUSES

F1 · F2 solenoids · F3 main 5 V rail

Ref	Protects	In line with	Rating	Part number	Size
F1	Winner-Lock solenoid	+5 V feed to J4	T2A slow-blow 250 V	Littelfuse 0218002.MXP	5 x 20 mm
F2	Coin-Lock solenoid	+5 V feed to J5	T2A slow-blow 250 V	Littelfuse 0218002.MXP	5 x 20 mm
F3	Main 5 V rail (inline)	5 V supply lead into J3	T3.5A slow-blow 250 V	Bussmann GDC-3.5A	5 x 20 mm

Holders: F1/F2 use Keystone 3517 board clips (2 each); F3 uses an inline 5 x 20 mm holder in the 5 V supply lead.
F1/F2 protect the two small 5 V solenoids (SparkFun 'Solenoid - 5V Small') - T2A slow-blow covers their inrush.
F3 guards the whole shared 5 V rail (lamps + LED strip + coils) right where the 5 V adapter feeds J3.
Slow-blow (T) rides out coil / LED inrush; any 250 V 5 x 20 mm part clears 5 V fine.

- Solder the four **Keystone 3517** clips — two per fuse — at the F1 and F2 positions. They only go one way; the fuse snaps in across the pair.
- Drop in the fuses: **F1** guards the Winner-Lock coil (J4), **F2** the Coin-Lock coil (J5). Both are **T2A slow-blow, 5×20 mm** (Littelfuse 0218002.MXP).
- Slow-blow (T) rides out the coil's turn-on surge; it opens only on a real short. Any 250 V 5×20 mm part is fine on the 5 V rail.

3c The inline main fuse

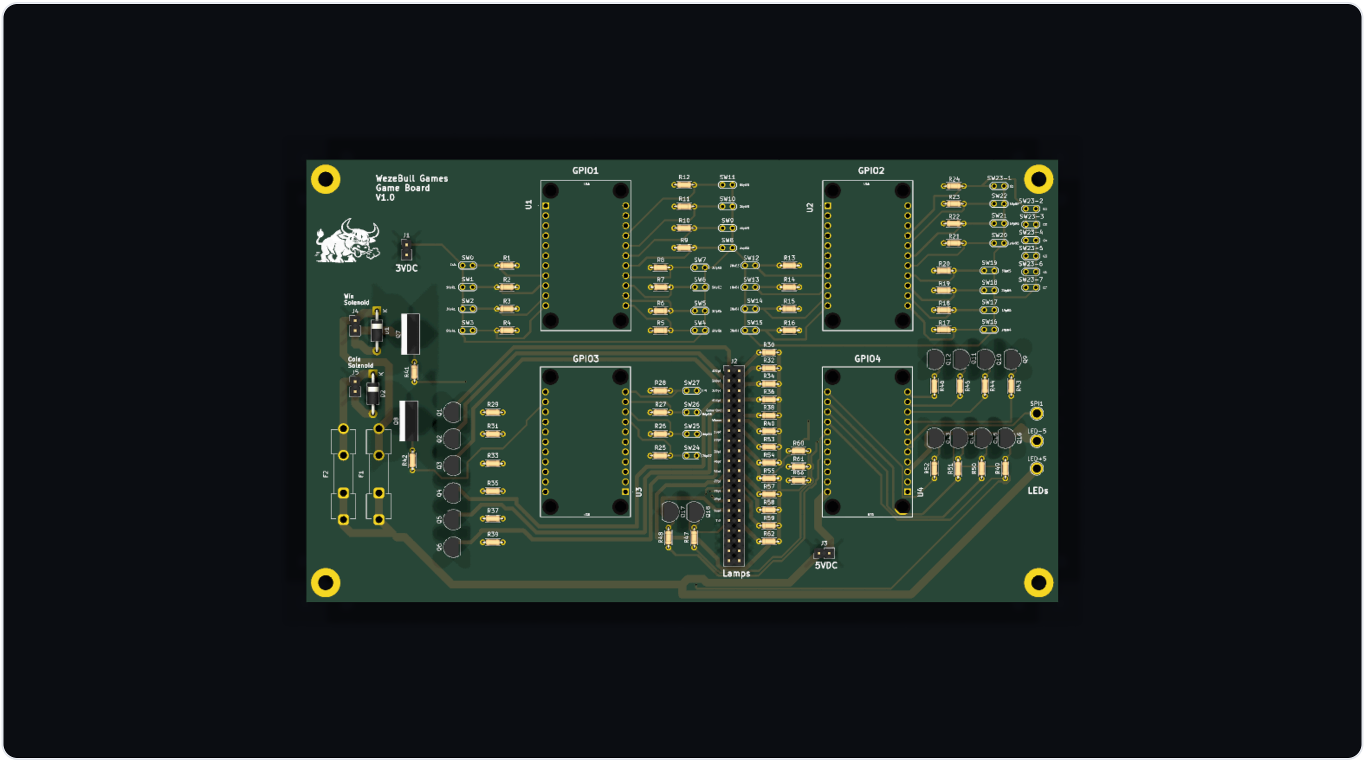
F3 — one fuse that guards the whole 5 V rail

- F1/F2 only protect the coils. **F3 protects everything on 5 V** — lamps, LED strip and coils — right where the adapter feeds the board.
- It isn't on the PCB: cut the **+5 V** wire coming from the adapter and splice the **inline holder** (uxcell 18 AWG, screw-terminal) into the break. Leave the ground/return wire whole.
- Fit a **T3.5A slow-blow, 5×20 mm** fuse (Bussmann GDC-3.5A). Sized above the full lamp + strip + coil draw, below the adapter's limit.

> Rule of thumb: F3 ≈ your steady current + inrush headroom, and always below what the wiring can carry. 3.5 A suits a ≥5 A adapter here.

4 Headers & sockets

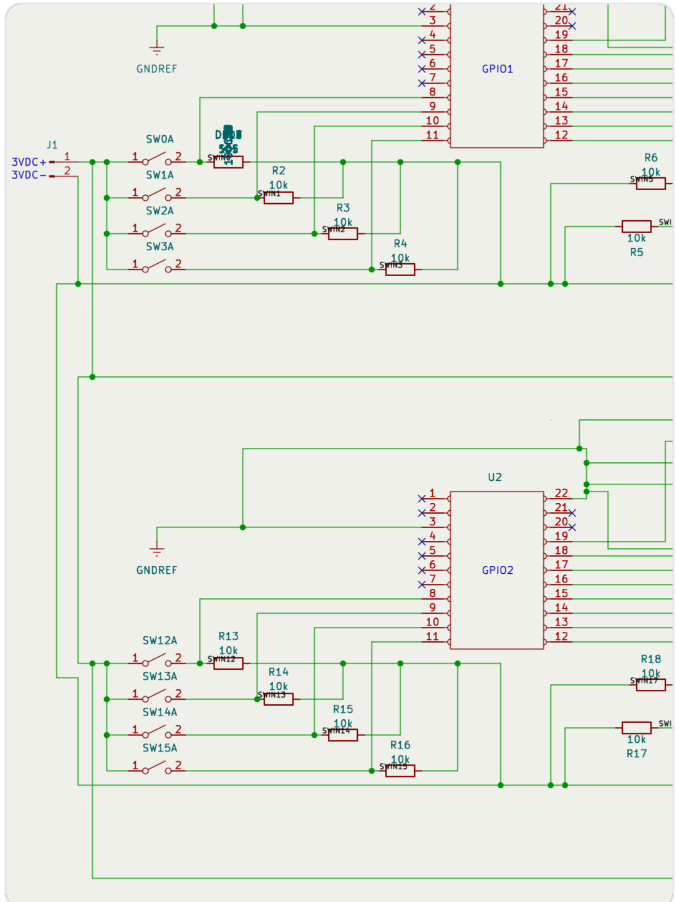
J1–J5, the J2 lamp header, and the 4 FT232H sockets



- Solder the **2×20 header (J2)** — the lamp/output connector — and the four **2-pin headers**: J1 (3.3 V in), J3 (5 V in), J4 (Winner-Lock), J5 (Coin-Lock).
- Tack one pin first, check the header sits flat and square, then solder the rest — a skewed 40-pin header is miserable to fix.
- Fit the four FT232H sockets with even spacing so the boards sit flat and don't touch. The footprints already have clearance; don't skew them.
- Note which socket is GPIO1–GPIO4 (silk) — it must match the serial you programmed into each board.

5 Wire the switches

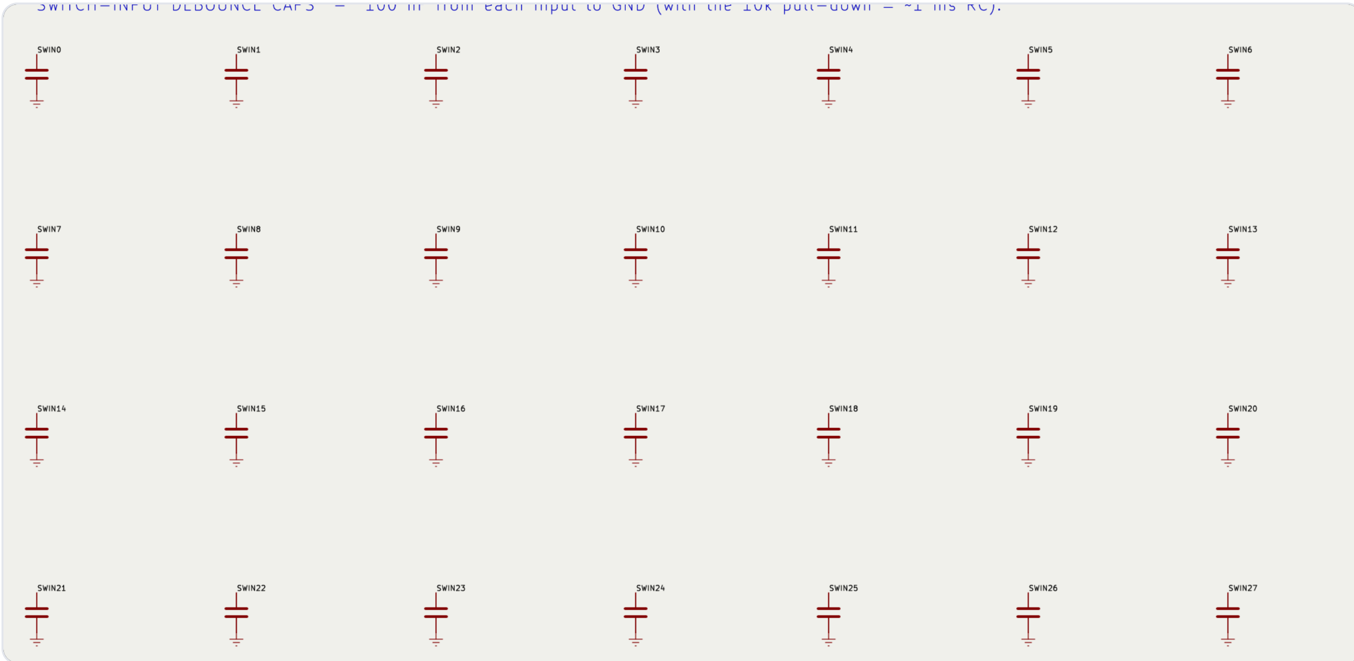
28 inputs → J1 commons, pulled to ground



- **28 inputs** across GPIO1–3 (D4–D7, C0–C7). One side of every playfield switch ties to the shared **J1 3.3 V common**; the other side goes to that switch's FT232H input pin.
- An on-board **10 kΩ pull-down** holds each input LOW; closing the switch pulls it to 3.3 V (a real HIGH) and the software reads the hit.
- **Keep the common at 3.3 V — never 5 V.** The FT232H inputs are **not** 5 V-tolerant; 5 V on an input can kill the chip.
- The **7 gobble holes** all report as one input (S23): wire their switches **in parallel** so any of them trips it.
- Run it as a harness (see the last pages): one 3.3 V bus wire daisy-chained to every switch, plus one signal wire per switch back to the board.

5b Fit the input caps

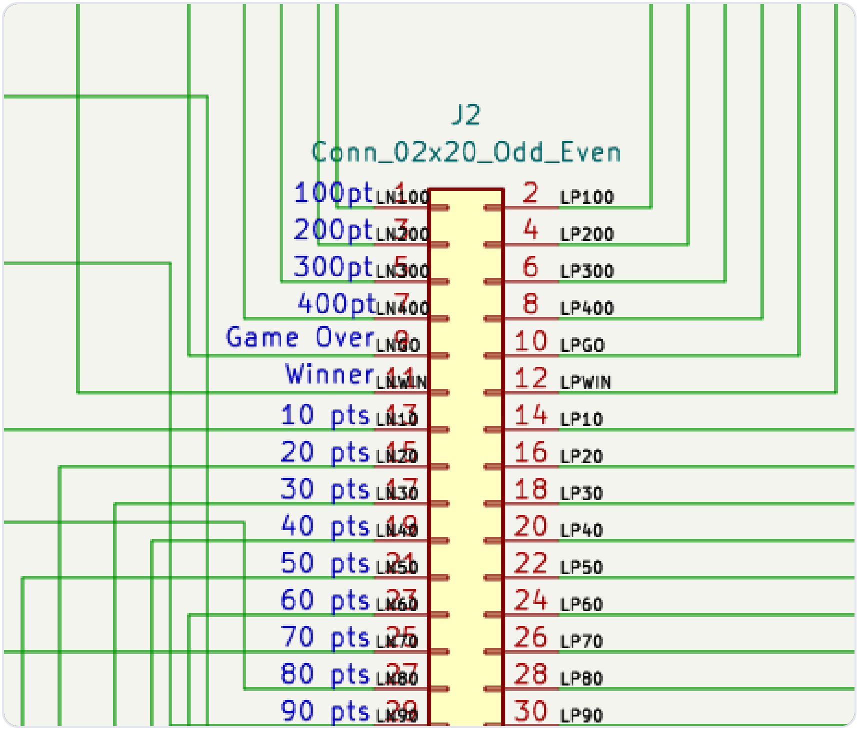
28× 100 nF on the board's back — debounce & noise



- Each switch input gets a **100 nF (0.1 µF) ceramic cap to GND** — C1–C28. With the 10 kΩ pull-down it forms a ~1 ms RC filter that kills contact chatter and noise picked up on the long playfield leads.
- They're **SMD 0805 on the back of the board**, one tucked next to each pull-down (GND pad on the ground pour, the other to the input). Solder them last.
- Optional but recommended — the game still works without them (the software also debounces), but they make the long switch runs far more reliable.

6 The score lamps

off-board #555 wedge LED bulbs on J2



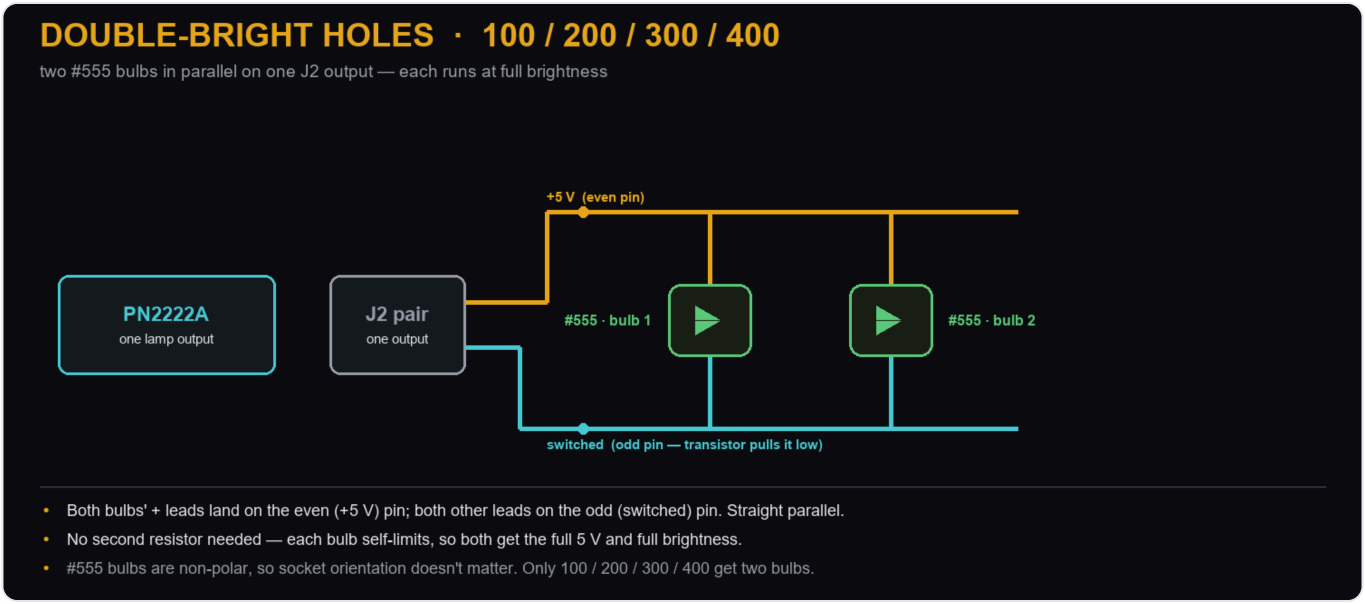
- Every score lamp — the tens **10–90**, the reels **100–400**, and **Winner / Game-Over / Tilt** — is a **#555 wedge LED bulb** in a twist-lock socket, off the board.
- Each bulb plugs across one **J2 pin pair**. The **odd** pin (left, labelled with the lamp name) is switched by the transistor; the **even** pin (right) is +5 V through that lamp's 0 Ω link. The labels on J2 tell you which pair is which lamp.
- Solder the socket's two leads to that pin pair. The #555 bulbs are **non-polar**, so which lead goes to which pin doesn't matter.

> Wire it as a harness (see the last pages): a 2×20 plug on J2, fanning out to all the sockets, is far tidier than 40 loose wires.

6b

Double up the big holes

100 / 200 / 300 / 400 — two bulbs each

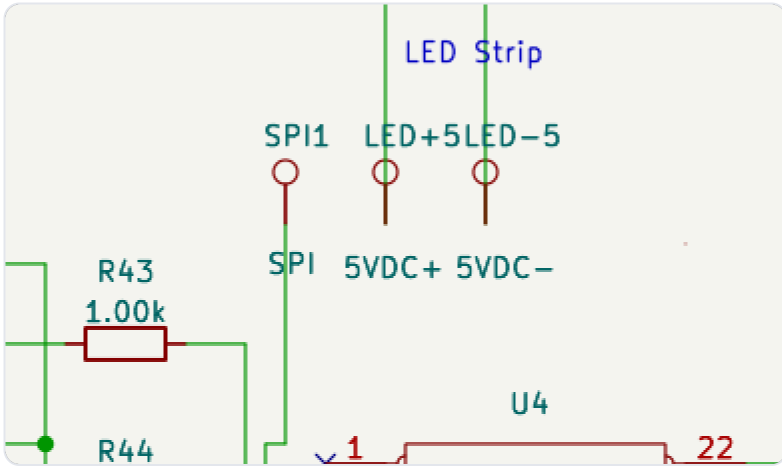


- The four high-score holes each drive **two** #555 bulbs for extra punch. Wire the second bulb **in parallel** on the same J2 pair as the first.
- Both bulbs' **+** leads land on the even (+5 V) pin; both other leads on the odd (switched) pin.
- No extra resistor — each bulb self-limits, so both run at full brightness. The transistor easily drives two.

6c

Coils & the LED strip

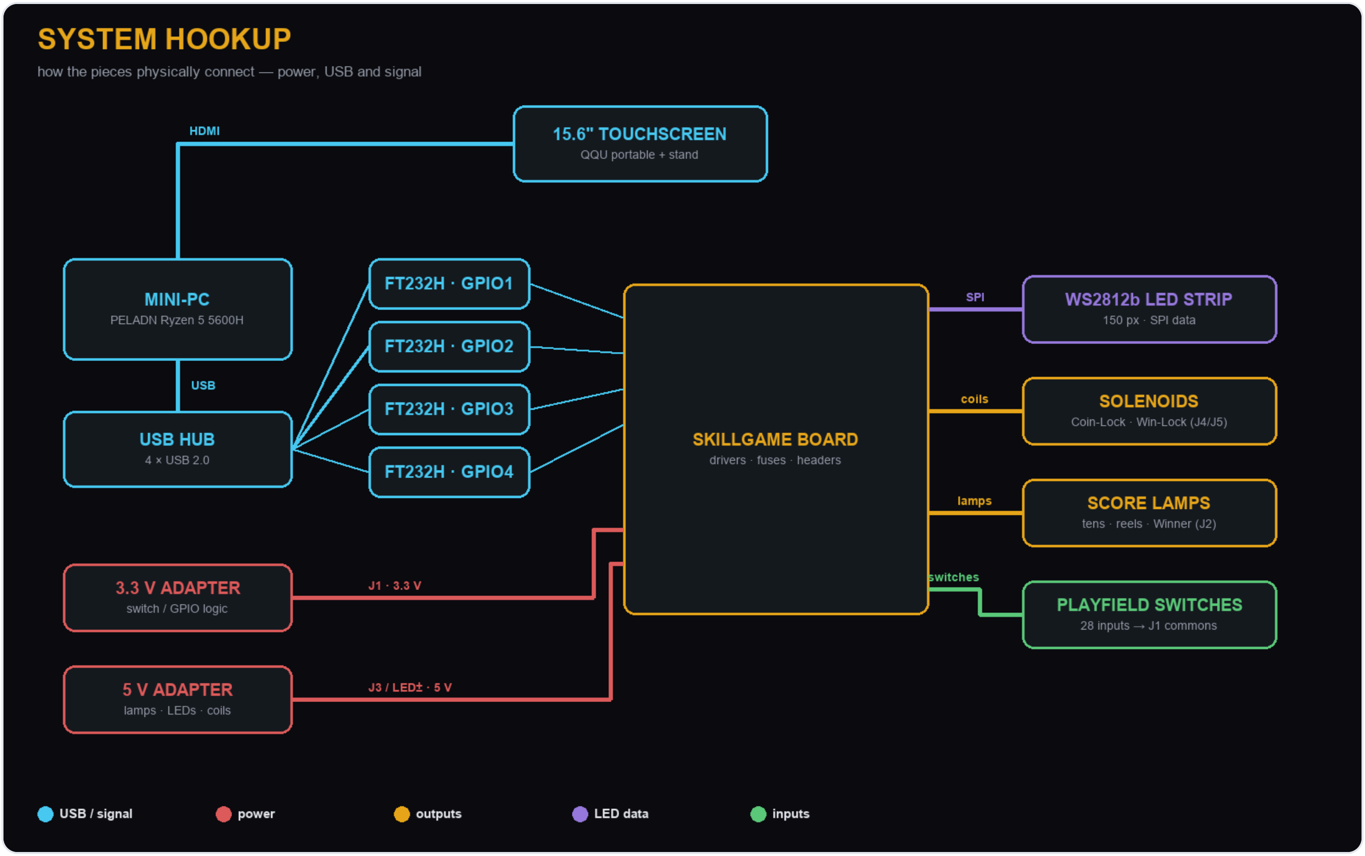
solenoids off J4 / J5, strip off SPI



- **Coin-Lock** → **J5**, **Winner-Lock** → **J4**. These are the fused (F1/F2), TIP120-driven 5 V coils — keep the flyback diode across each coil.
- LED strip **data** → **GPIO4's D1** (SPI MOSI). Data is one wire plus a common ground.
- Power the strip from its **own 5 V feed** (LED+5 / LED−5), **not** J3, so its heavy draw can't sag the lamp/logic rail. Tie all grounds common.

7 Hook up the system

PC · hub · monitor · power



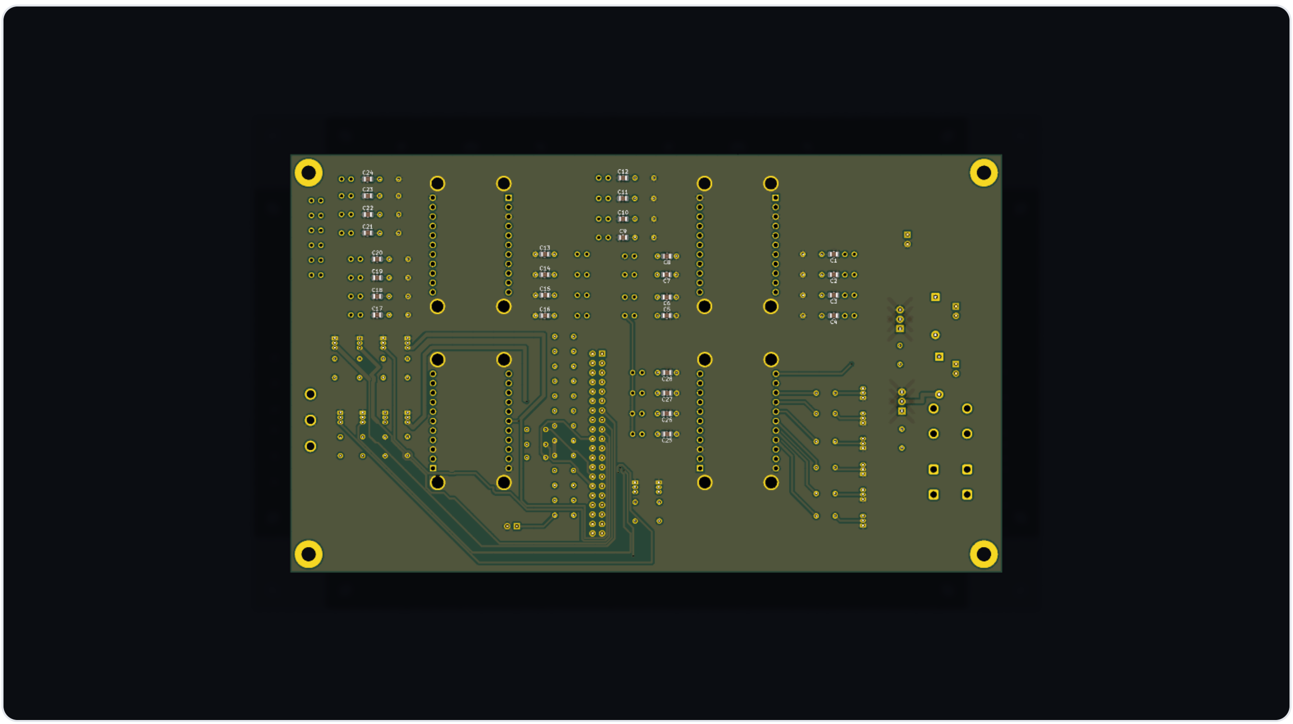
- **Mini-PC → HDMI → monitor** (the QQU 15.6" touch panel), and its USB-C touch lead back to the PC.
- **Mini-PC → USB hub → the four FT232H boards** — one hub port per board.
- **3.3 V adapter → J1** (switch commons). **5 V adapter → J3** (through the F3 inline fuse) and to the LED strip's own **LED± feed**.
- **Tie every ground together** — the 3.3 V return, the 5 V return, the LED strip and the boards all share one common ground, or nothing reads right.
- Double-check J1 is 3.3 V and J3 is 5 V before powering on. They are not interchangeable.

8 Power on & self-test

let the software check your wiring

- Launch SkillGame, open **Diagnostics**, and run **Self-Test** — it lights every lamp, pulses the coils and sweeps the LED strip.
- Use the **Switch Monitor** to press each playfield switch and confirm it registers (and clears).
- Any input stuck HIGH shows as a fault ring — check that switch's wiring and pull-down.
- Green badge on Diagnostics = all boards present and healthy. You're done. **ALL HUFF · NO BULL.**

> The board back is on the next page for reference, followed by the full schematic.



Wiring harnesses

THE CABLES THAT JOIN THE PLAYFIELD TO THE BOARD

- The board only holds the transistors, resistors and connectors. The **lamps** (backglass) and **switches** (playfield) live off-board, so two harnesses connect them:
- **Lamp harness** — a 2×20 plug on J2 fans out to the 20 #555 wedge sockets.
- **Switch harness** — one 3.3 V common bus to every switch, plus a signal wire from each switch back to the board.
- Plus the **inline fuse cable** (F3) spliced into the 5 V feed.
- Each cable is shown three ways on the next pages: flat with every solder point, a 3D view, and **where every point lands on the real game**.

L

Lamp harness — wiring

J2 plug → 20 #555 sockets, every solder point

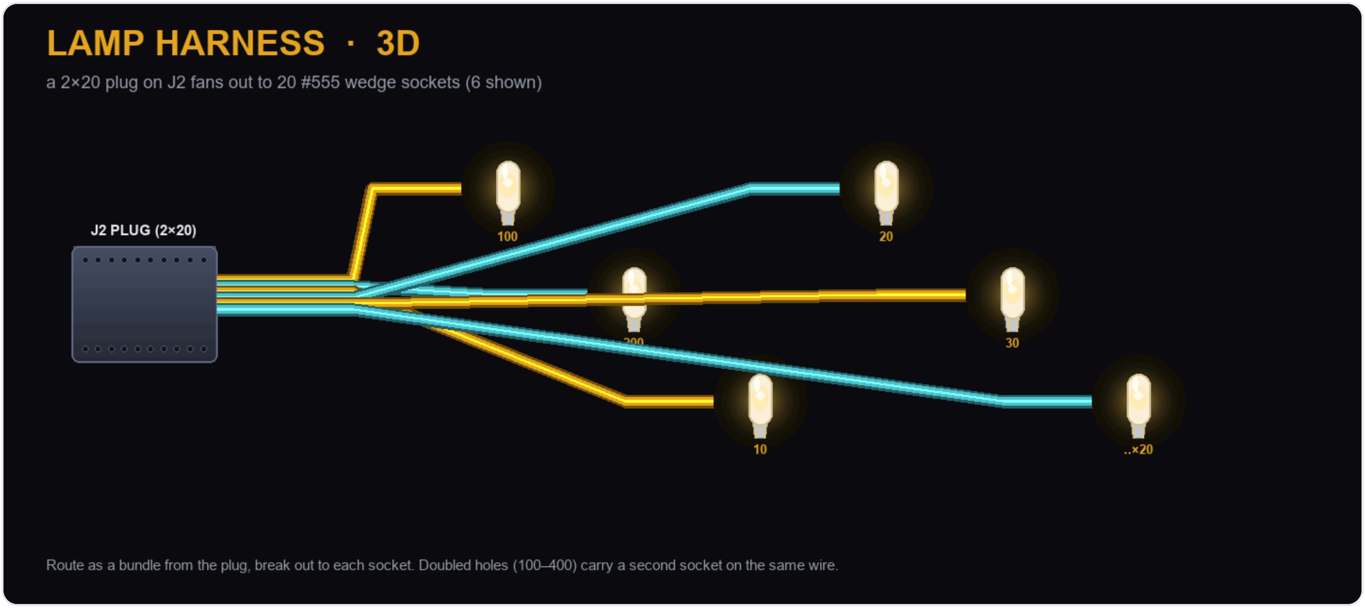
LAMP HARNESS

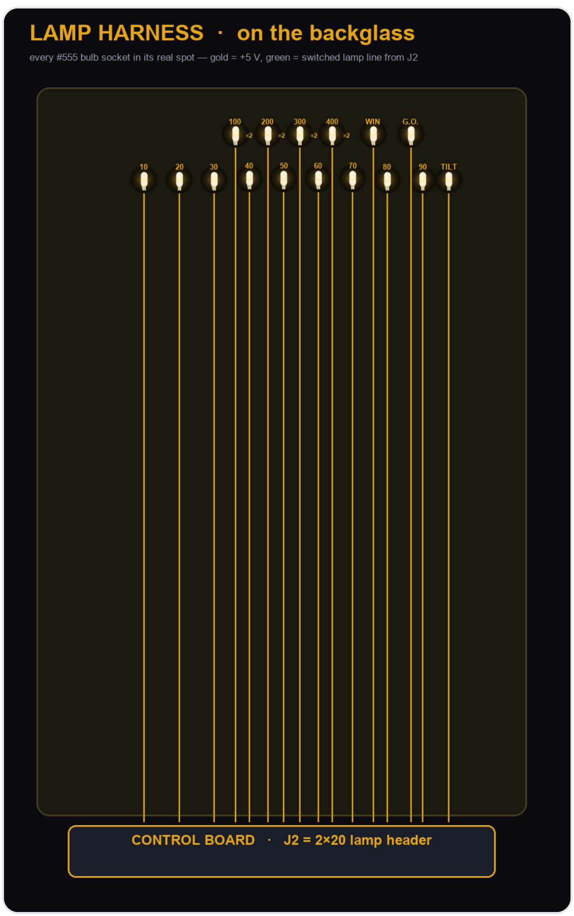
one 2×20 plug on J2 → 20 #555 wedge sockets · gold = +5 V (even pin), cyan = switched (odd pin)

- Solder each socket's two leads to its J2 pin pair (dots). 100 / 200 / 300 / 400 carry a second socket on the same pair.
- GROUND closes ON THE BOARD — the driver transistor's emitter to GND. Only 2 leads run to each socket, no ground wire.
- Non-polar #555 bulbs — either lead to either pin. Keep +5 (gold) and switched (cyan) consistent so the harness stays readable.

L Lamp harness — 3D & where it lands

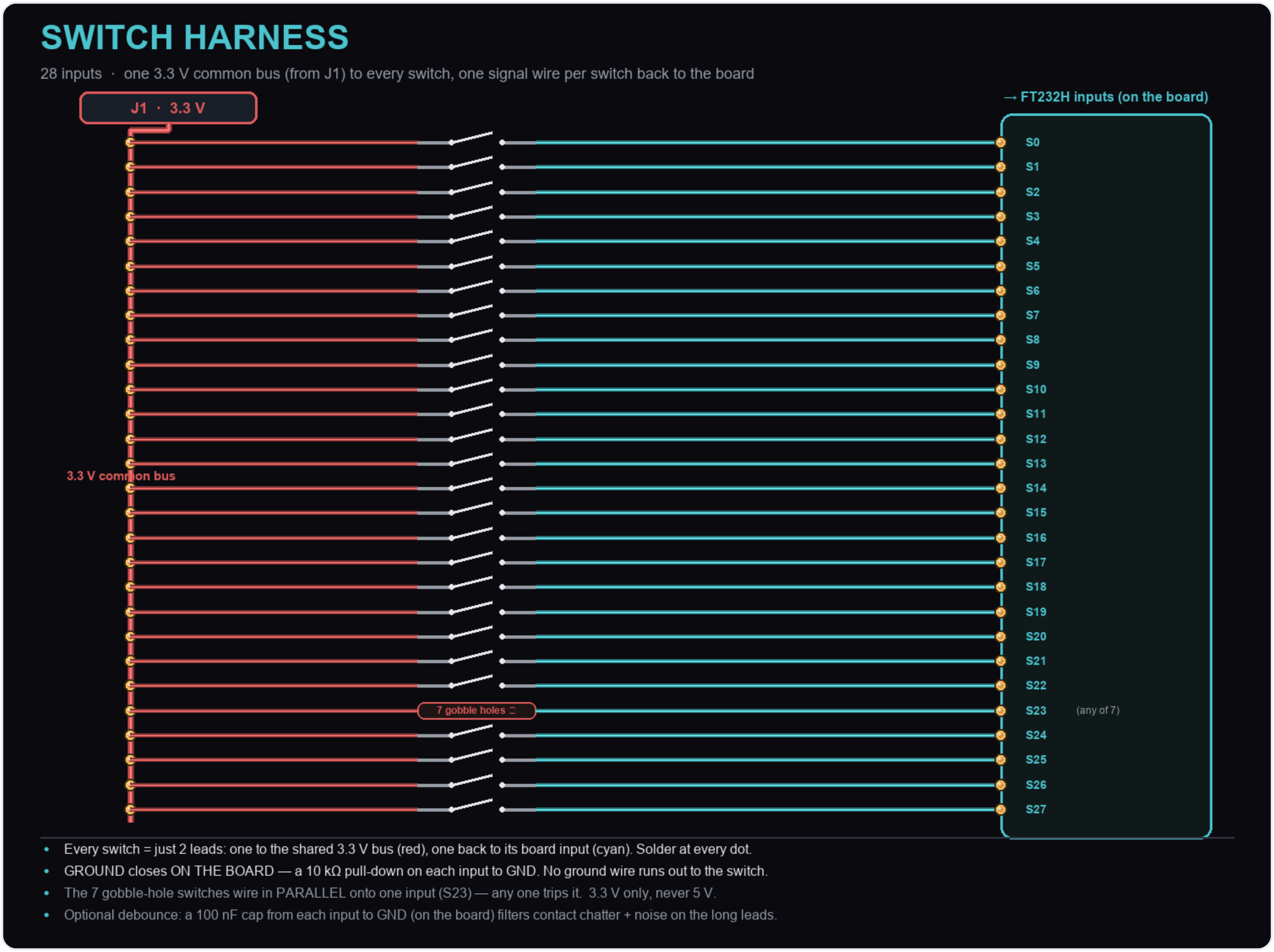
the physical cable, and each bulb's spot on the backglass





Switch harness — wiring

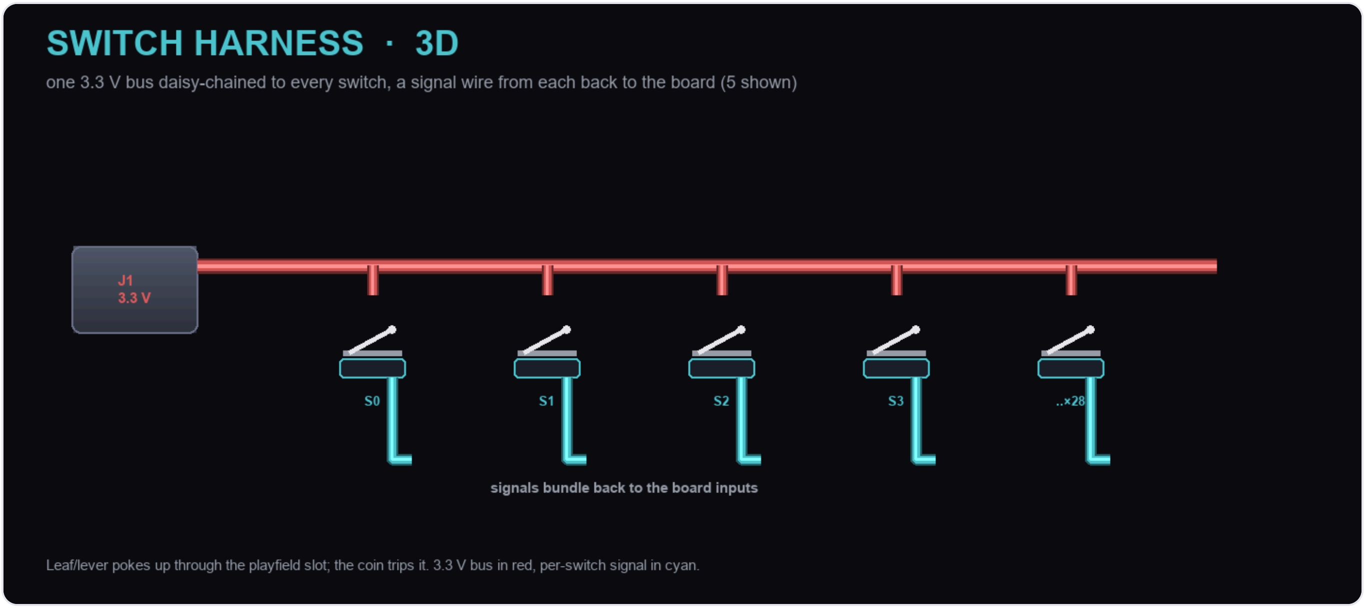
3.3 V bus + one signal per switch, every solder point

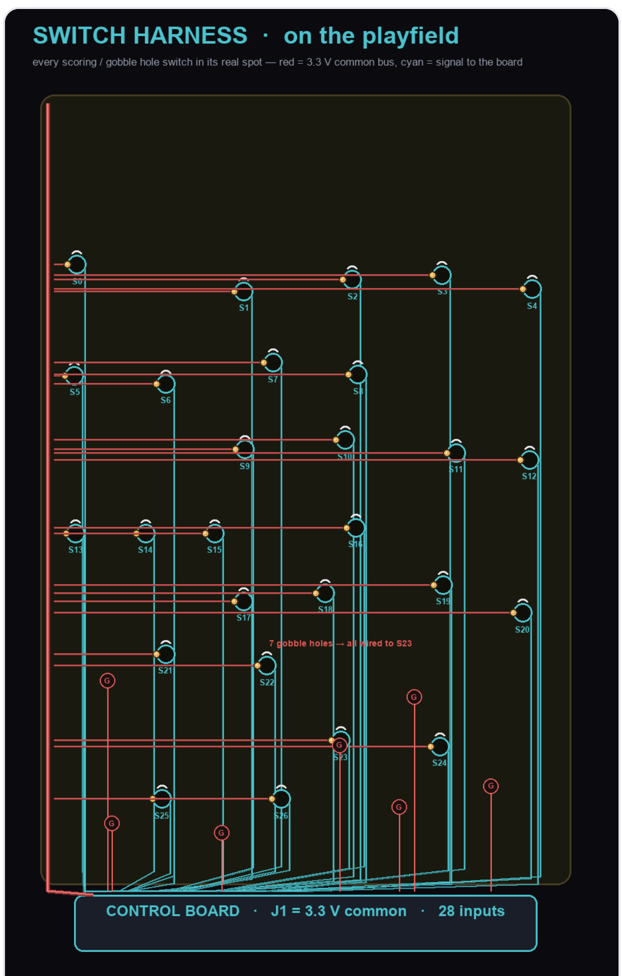


Which switch? The original Bally rollover-wire leaf switch (ASW-A1-152) is **discontinued**. Use a **mini snap-action microswitch with a hinge lever** — the lever pokes up through the playfield slot and the coin trips it (~$0.80 ea, in stock). Pinball-authentic alt: Action Pinball **SW-1A-120** ($8.39, in stock) with a rollover wireform. Best of all — reuse your machine's originals.

Switch harness — 3D & where it lands

the physical cable, and every switch's spot on the playfield





Inline fuse cable — 3D

F3 in the 5 V feed to J3

Full schematic

REFERENCE